Xiaoyu Yan

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Education

Northwestern University — Evanston,IL
Ph.D. Transportation System Analysis and Planning — *Expected Jun. 2028*

University of Illinois at Urbana-Champaign — Urbana, IL
B.Sc. Civil Engineering, Summa Cum Laude, GPA:3.91/4 — *May. 2023*

Zhejiang University — Zhejiang, China
B.Eng. Civil Engineering, GPA:3.98/4 — *Jun. 2023*

Technical Skills

Data Science: statistical modeling, predictive modeling, Bayesian modeling, experimentation design, causal inference, machine learning, deep learning (PyTorch, scikit-learn), natural language processing
AI Agent: agentic AI, multi-agent LLM, RAG, SFT, post-training, tool-use agent, synthetic data generation
Language & Tools: Python, SQL, R, C++, JavaScript, Git, MATLAB, Docker, LaTeX
Mobility Analytics: optimization, ArcGIS, fleet operations, travel demand modeling, marketplace

Experience

TransUnion — Chicago, IL
Data Science and Analytics Intern — *Jun. 2026 – Sep. 2026*

- Built an end-to-end data pipeline on **Google Cloud Platform**, processing **1.4M+ housing records** using **SQL** and **R** for data extraction, validation, and feature engineering
- Benchmarked **TabPFN** against XGBoost for eviction risk prediction, achieving **50% higher KS** and **3× lift** in the top 20% risk segment, while optimizing **GPU utilization, runtime, and accuracy** trade-offs
- Developed an automated model evaluation and retraining framework incorporating **SHAP analysis** and **population stratification** to investigate heterogeneous risk factors and potential confounding effects

Northwestern University | Center for Connected and Automated Transportation — Evanston, IL
Graduate Researcher — *Sep. 2023 – Present*

Agentic AI for Automated Transportation Policy Making

- Built an agentic **LLM-driven multi-agent platform** to simulate citywide transit policy referenda, analyzing voting outcomes and heterogeneous community preferences across cities
- Integrated a **utility-based transit design model** to benchmark policy outcomes and developed **statistical models** to investigate LLM voting mechanisms and policy alignment
- Applied the **Plackett–Luce model** to estimate community preferences and developed a **regret-guided adaptive policy search** to balance policy performance and public acceptance

Transportation Insecurity Inference Using Large-Scale Mobile Phone Location Data

- Processed **4.6 million GPS records** using **Python** and **SQL**, extracting spatiotemporal features and applying **hierarchical clustering** to identify mobility patterns, improving classification accuracy by 8%.
- Built a **semi-supervised learning pipeline** for transit/driving classification, achieving **96% accuracy**
- Built a **transfer learning framework** with GPS data to predict transportation insecurity across Detroit and Chicago, achieving 0.742 ROC-AUC on held-out survey data

Impact of Modularized Autonomous Vehicles on Transit System Design and Operations

- Built a stylized, **joint optimization** model for the fixed-route transit and paratransit operations for resource allocation, delivering policies that minimize total travel time within budget constraints
- Stress-tested fleet plans for Chicago by calibrating parameters and running sensitivity analyses on vehicle capacity and multi-period demand, **identifying fleet operation plans** and time-cost trade-offs for **planning decisions**

Alibaba-Zhejiang University Joint Research Institute — Zhejiang, China
Applied AI Scientist Intern | Institute of Frontier Technologies — *Sep. 2021 – Jun. 2023*

- Built a city-scale **trip-chain reconstruction** pipeline from sparse GPS data with DBSCAN and Random Forest
- Developed a **multimodal travel-mode identification** workflow with feature engineering and AdaBoost classifier, achieving strong performance across modes and **93.5% metro identification accuracy**
- Implemented a link-level CO_2 accounting framework with sample expansion and emission-factor modeling, and built a **cloud-backed interactive platform** for real-time, multi-scale emissions **monitoring over 9.19M records**

Projects

Agentic AI for Personalized Trip Planning and Booking Optimization *Nov. 2025 – Present*

- Design an **LLM-based multi-agent framework** with **agent orchestration**, integrating ticket search, active preference elicitation, and memory-based utility learning for personalized trip planning
- Develop a **natural language-to-optimization pipeline** for **structured constraint extraction** and **optimization solver integration**, enabling automated, constraint-aware itinerary generation

Fairness-Enhanced Deep Learning in Travel Demand Prediction *May. 2023 – Sep. 2023*

- Built a fairness-aware **spatiotemporal neural network** (STGCN) for Chicago travel-demand forecast
- Added an equality-enhancing loss to reduce demographic/spatial disparity, **improving fairness up to 48%**

Travel Behavior Study on Citi Bike Users in New York City *Jun. 2022 – Aug. 2023*

- Analyzed NYC Citi Bike data to quantify user behavior shifts across COVID with **statistical validation**
- Built a station-level clustering pipeline with **Gaussian Mixture Model** and fused Points-of-Interest data to distinguish commuting vs. leisure clusters and translated findings into operational strategies

Leadership

President | Northwestern Transportation Club *Sep. 2025 - Present*

- Organize monthly events for 50+ members; manage budget applications and account reconciliation
- Forge the partnership with industry, and coordinate site visits, e.g. Amazon, Chicago DOT, and CTA, etc

Board Member | ASCE Student Chapter at Zhejiang University *Jun. 2022 - Jun. 2023*

- Organized and hosted professional seminars Civil Talk and campus-scale structure competitions

Honors & Awards

- **TYLin WTS Chicago President's Scholarship**, WTS Greater Chicago Area Chapter *2026*
- **Cabell Fellowship, Northwestern University** (Selected as one of 10 fellows university-wide) *2023*
- **Outstanding Graduate Award, Zhejiang Province & Zhejiang University** (Top 3%) *2023*
- **China National Scholarship** (Top 1% nationwide) *2022*
- **Dean's List, University of Illinois Urbana-Champaign (UIUC)** *2022*

Publications & Conferences

[1] **Yan, X.**, Dai, T., Nie, Y. Incorporating Community Feedback into Transportation Policy Making through an Agentic AI framework. Presented at 2026 Informs Annual Meeting, 2026.

[2] **Yan, X.**, Dai, T., Nie, Y. Addressing the alignment problem in transportation policy making: an LLM approach. *Transportation Research Part C: Emerging Technologies*, 188, 105686. Earlier version presented at ***NeurIPS** 2025 Workshop Responsible and Trustworthy Foundation Models.*

[3] Dai, T., **Yan, X.**, Kivestu, P., Chen, Y., Murphy, A., Nie, Y. Learning Transportation Insecurity from Mobile Phone Data. presented at *Transportation Research Board 106th Annual Meeting.*

[4] **Yan, X.**, Zheng, H., Nie, Y. Joint design of fixed-route and paratransit services with autonomous pods. presented at the *Transportation Research Board 104th Annual Meeting*, Washington, D.C. Presentation Number: TRBAM-25-05686, 2025.

[5] Ariana, D.,**Yan, X.**, Chen, Y. A Two-Layer Mobility Service Index: A Framework for Multidimensional Urban Mobility Assessment. under review for *Transportation Research Board 106th Annual Meeting.*

[6] Dong, Z., Chen, C., Ouyang, J., **Yan, X.**, Liao, C., Lee, D., Chen, X. Understanding commuter preferences for on-demand shared autonomous electric vehicles. *Transportation Research Part D: Transport and Environment*, 140, 104621.

[7] Wang, K., **Yan, X.**, Zhu, Z., & Chen, X. M. (2024). Understanding bike-sharing usage patterns of members and casual users: A case study in New York City. *Travel Behaviour and Society*, 36, 100793.

[8] Liu, J., Li, J., Chen, Y., ..., **Yan, X.**, ..., Chen, X. (2023). Multi-scale urban passenger transportation CO_2 emission calculation platform for smart mobility management. *Applied Energy*, 331, 120407.

[9] Lai, D., Lin, L., **Yan, X.**, Li, Z., Xu, K., Demartino, C., Xiao, Y. (2021). Development of a Steel Fiber-Reinforced Rubber Concrete for Jacket of Bridge Piers Against Vehicular Impacts: Preliminary Results. In *International Conference of the European Association on Quality Control of Bridges and Structures* (pp. 1144–1151). Springer, Cham.